Lablup Track Challenge

JUNCTIONX Korea 2026

lablup

August 2026

Introducing Lablup

Democratizing [intelligence] for humanity

lablup

We're **working toward a world** where **intelligence** flows **like electricity**, reaching wherever it's needed.

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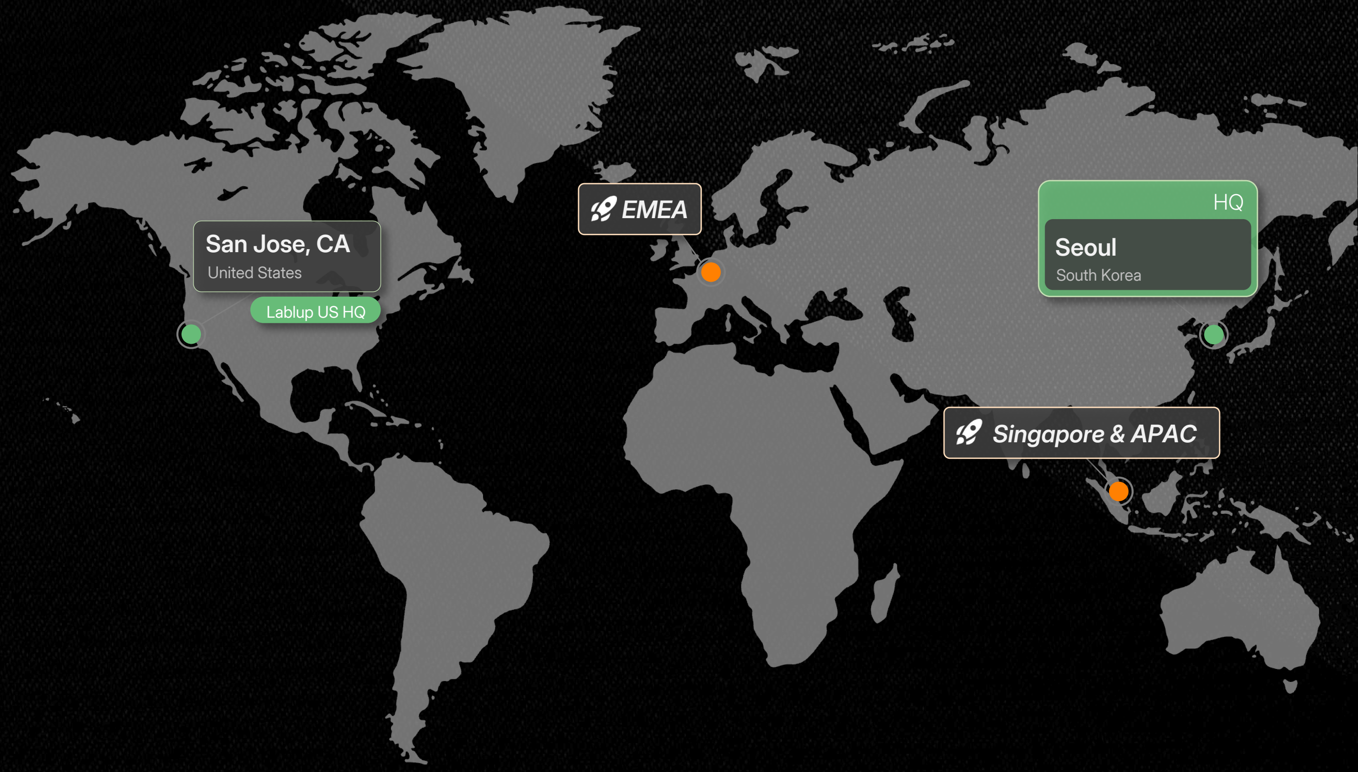
About Lablup

Making AI Accessible, Scalable, and Composable

Founded **2015.04** | Employees **46**

Korea's Only Profitable AI Infra-software Company

Developing platform that powers AI from its backbone



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Open-source native

Open-source is our foundation

Proud Sponsor of

CLOUD NATIVE COMPUTING FOUNDATION

Agentic AI Foundation

THE LINUX FOUNDATION

python SOFTWARE FOUNDATION

PyTorch Foundation

Customers

Powering customer success

SAMSUNG
LG Electronics
SMC SAMSUNG MEDICAL CENTER
CNU HOSPITAL
LIG Defense & Aerospace
HYUNDAI MOBIS
CJ

SAMSUNG WELSTORY
BANK OF KOREA
Shinhan Bank
AICA Artificial Intelligence Industry Cluster Agency
대한민국해군 REPUBLIC OF KOREA NAVY

ETRI 한국전자통신연구원
KISTI www.kisti.re.kr Korea Institute of Science and Technology Information
KIST Korea Institute of Science and Technology 한국과학기술연구원
HIRA HEALTH INSURANCE REVIEW & ASSESSMENT SERVICE
KOREAN AIR
42dot
JTS Jasmine Technology Solution

kt
kt cloud
NHN CLOUD
Samsung Research
경기도 GYEONGGI-DO

USC Center for Advanced Research Computing
Enabling scientific breakthroughs at scale
1398 성균관대학교 SUNGKYUNKWAN UNIVERSITY
KMU 국민대학교

Product

backend AI

The vendor-agnostic OS for every AI accelerator

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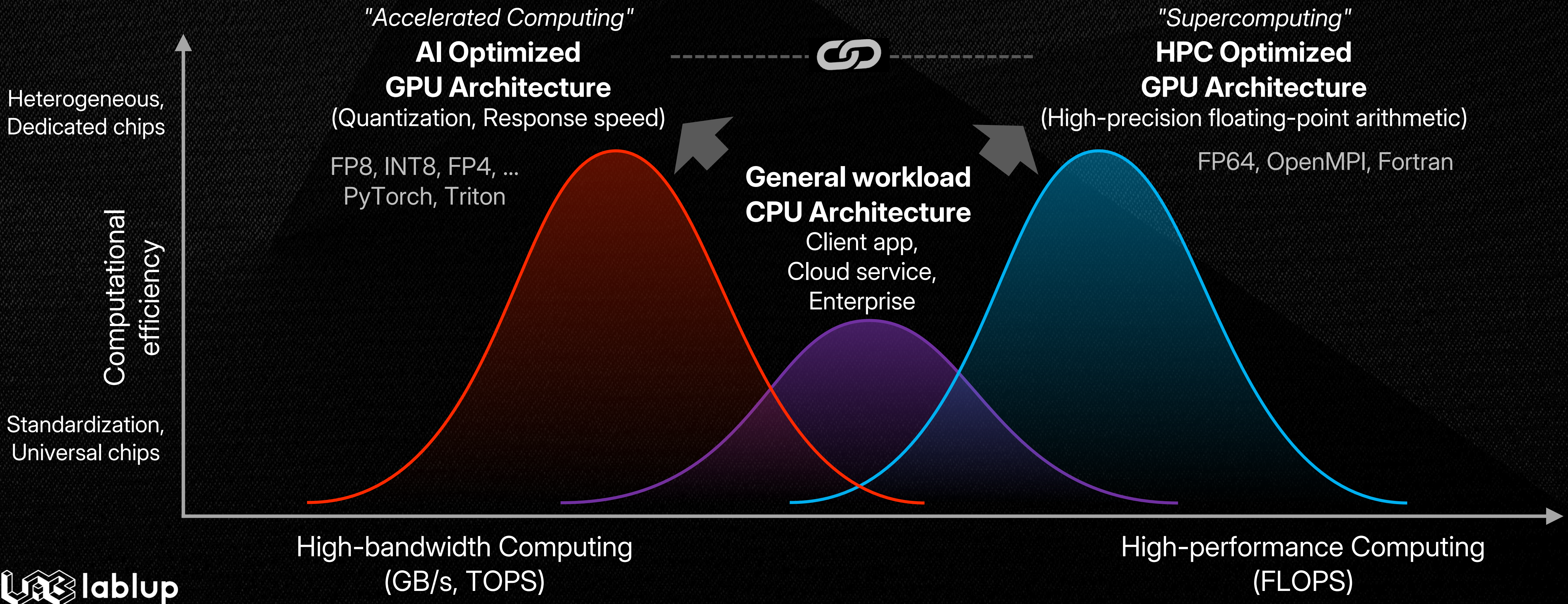
The direction of computing development

From HPC to AI, General-purpose to Accelerated



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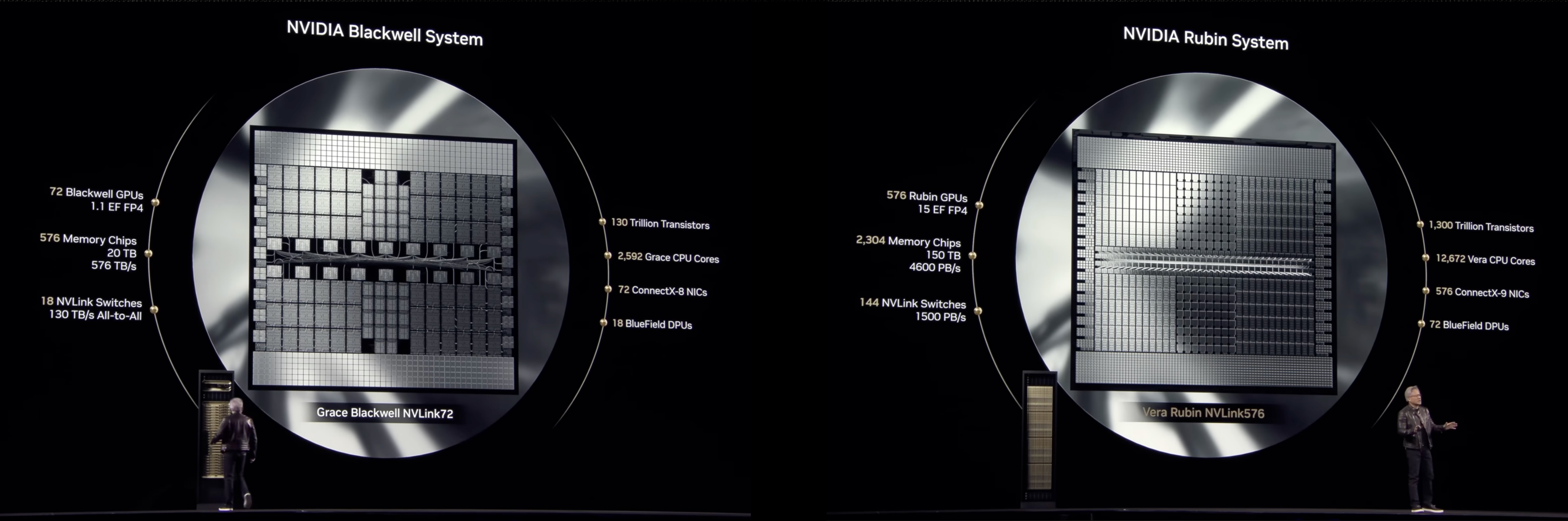
The direction of computing development

The Era of High-density, High-integration Accelerated computing



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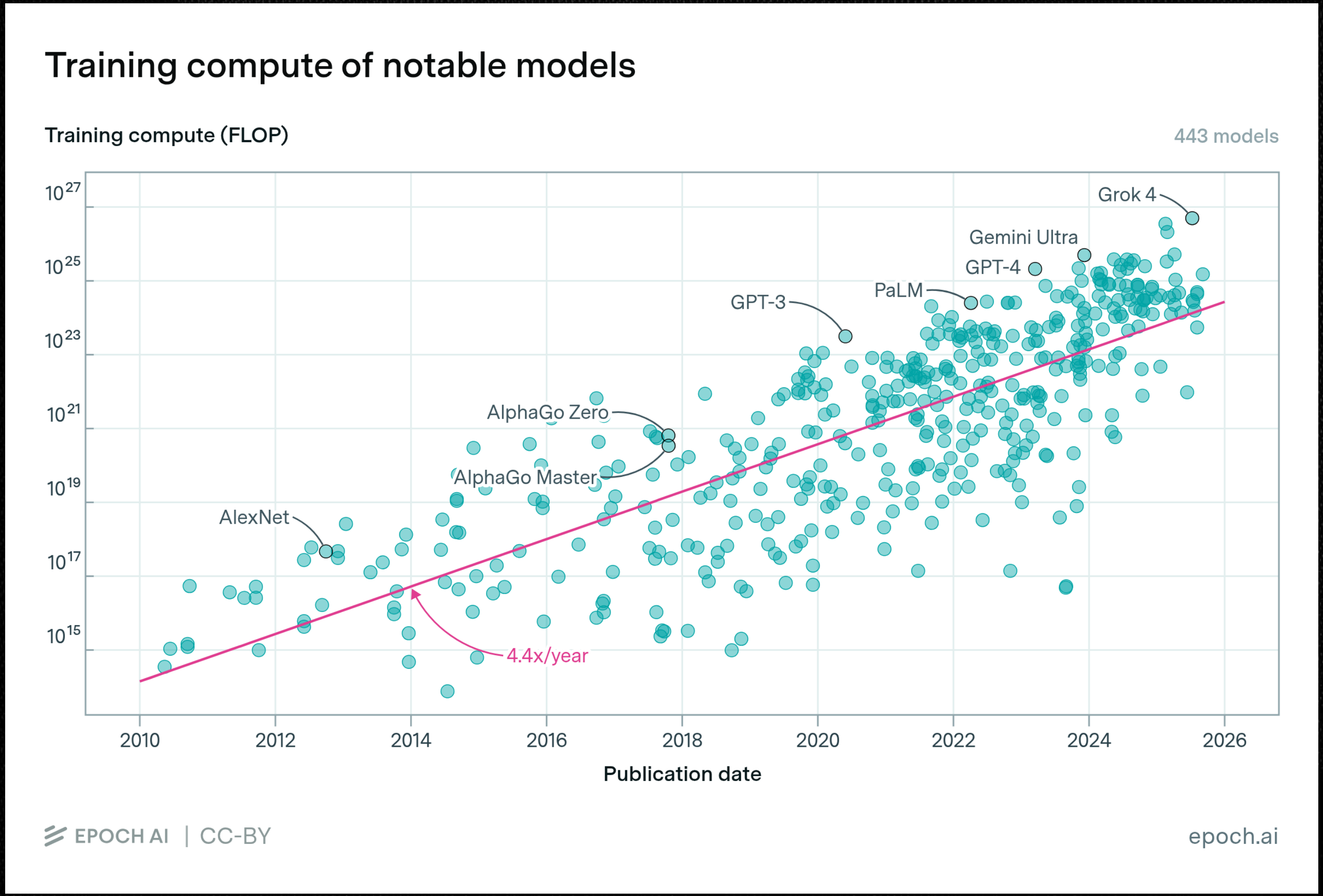
The challenge toward intelligent infrastructure

Infrastructure: the foundation of an intelligent future

Increased computing needs

4x/year

(Approx.)



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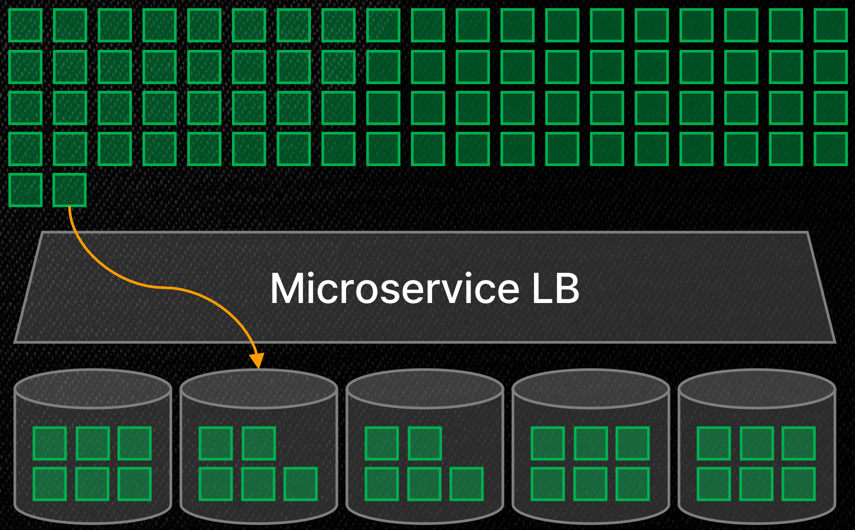
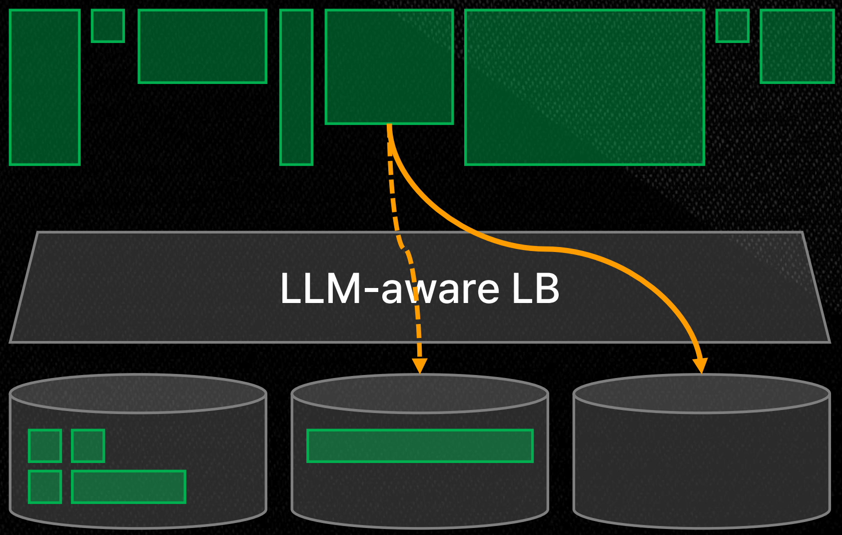
The challenge toward intelligent infrastructure

Technical difficulty in Building cost-effective LLM services

- General HTTP Workloads (Microservice)
  - Fast, uniform, affordable
  - Easy Preemption, simple scale-in & out



- LLM Service Workloads
  - Slow, non-uniform, high cost
  - Input-dependent context, node state-aware routing, scaling

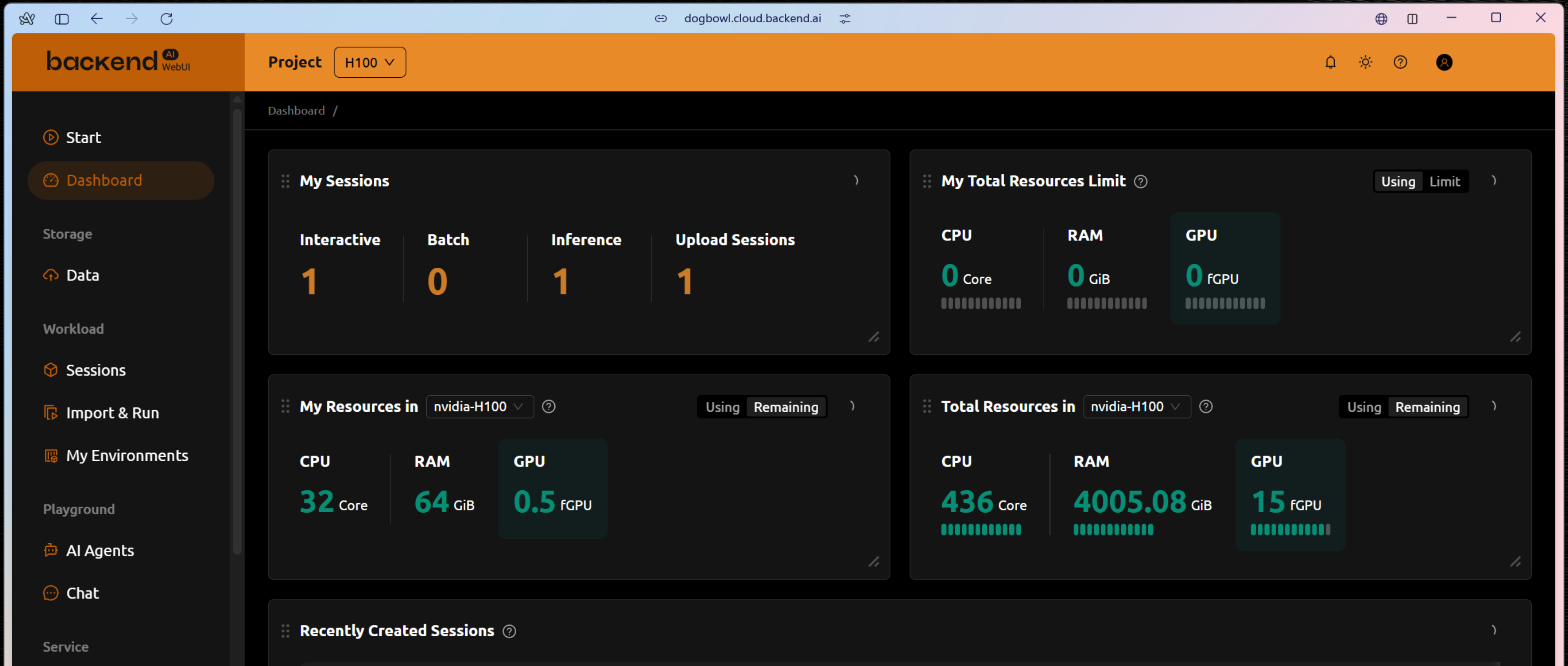


[1] LLM-d. JAX/OpenXLA DevLab (2025)

backend AI

[bæk ˈɛnd dɒt eɪ-aɪ]

The ‘AI OS’ for every infrastructure



backend AI

Scale GPUs your way

Slice a GPU into multiple virtual GPUs with **container-level GPU virtualization**

Accelerate your data and storage

RDMA and **NVIDIA Magnum IO GPUDirect Storage** support

Leverage heterogeneous AI accelerators

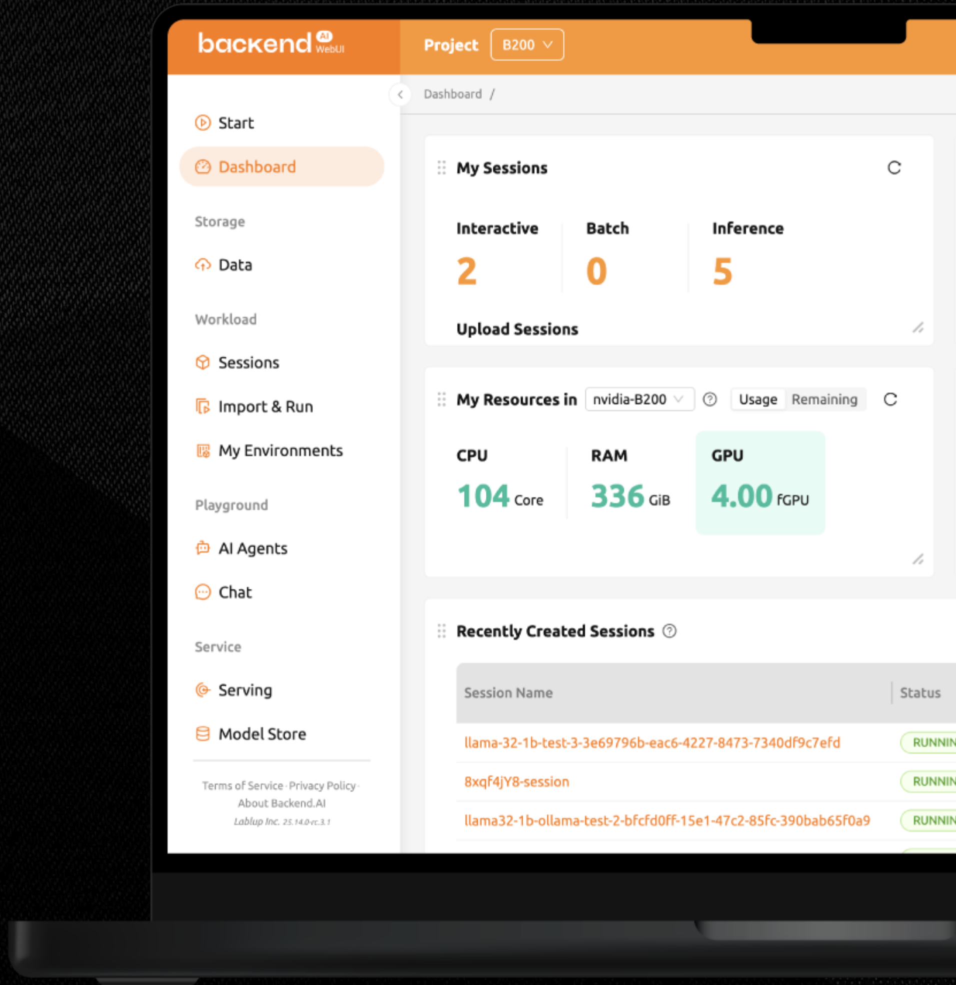
Supporting both **x86-64** and **aarch64**

Deploy confident in multiple environments

Operate on **air-gapped**, **on-premises**, **cloud**, even **hybrid** setups

Orchestrate Large-Scale AI Workloads

Manage **multi-tenancy**, **multi-node** workloads with Sokovan™ orchestrator



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backend AI

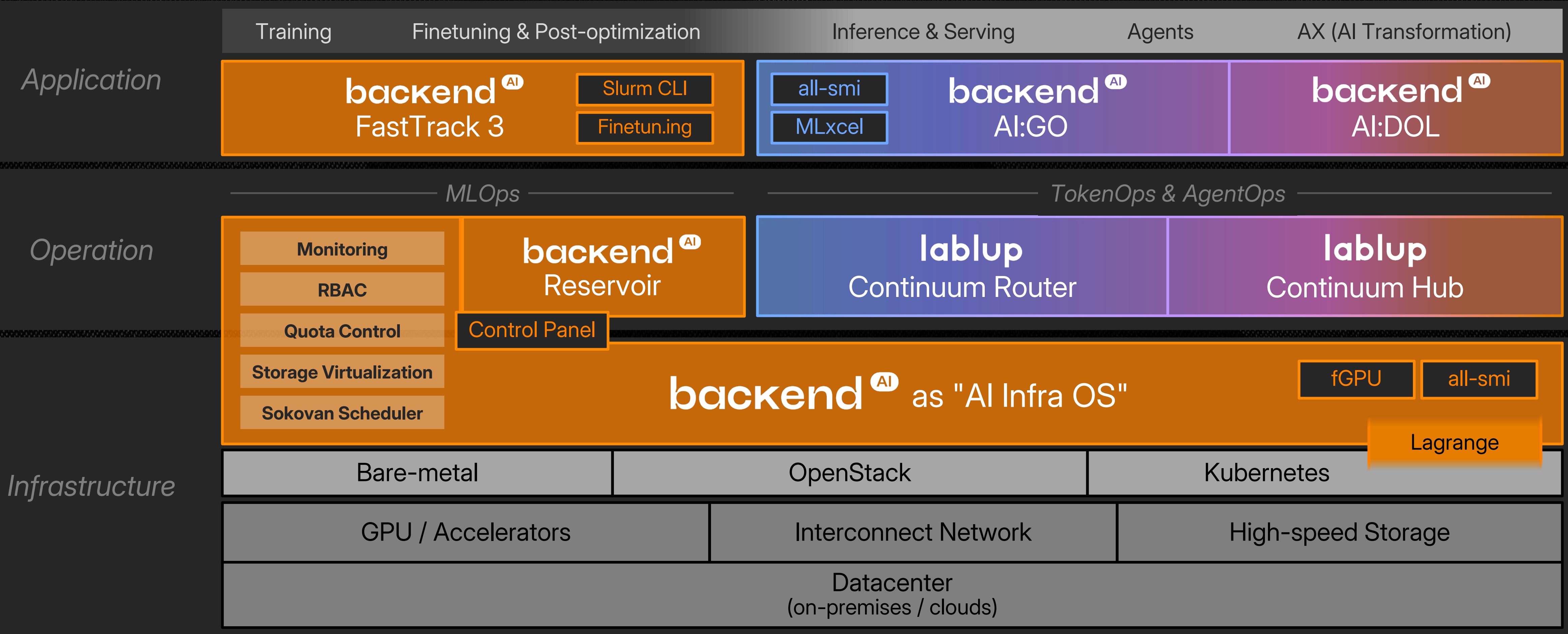
Proven on heterogeneous accelerators

NVIDIA Blackwell B300, B200 ...	**NVIDIA Jetson** TX, Xavier, Orin, Thor...	**NVIDIA Grace Blackwell** GB300, GB200, GB10 ...	**NVIDIA Hopper** H200, H100 ...
NVIDIA Ampere A10, A40, A100 ...	**NVIDIA Turing** Titan RTX, RTX 8000, T4 ...	**NVIDIA Ada Lovelace** L40S, L4 ...	**Groq** Groq card ...
Intel Gaudi Gaudi 2, 3, Arc Pro ...	**AMD Instinct** MI250, MI300 ...	**Furiosa** Warboy, RNGD ...	**Rebellions** ATOM, ATOM+, Rebel ...

And more (Graphcore, Sambanova, HyperAccel, Groq, etc.)

Vertical integration of AI infrastructure

Turn-key AI Enterprise Solution



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Continuum suite

Continuum Router

Enterprise-grade Multi-LLM gateway

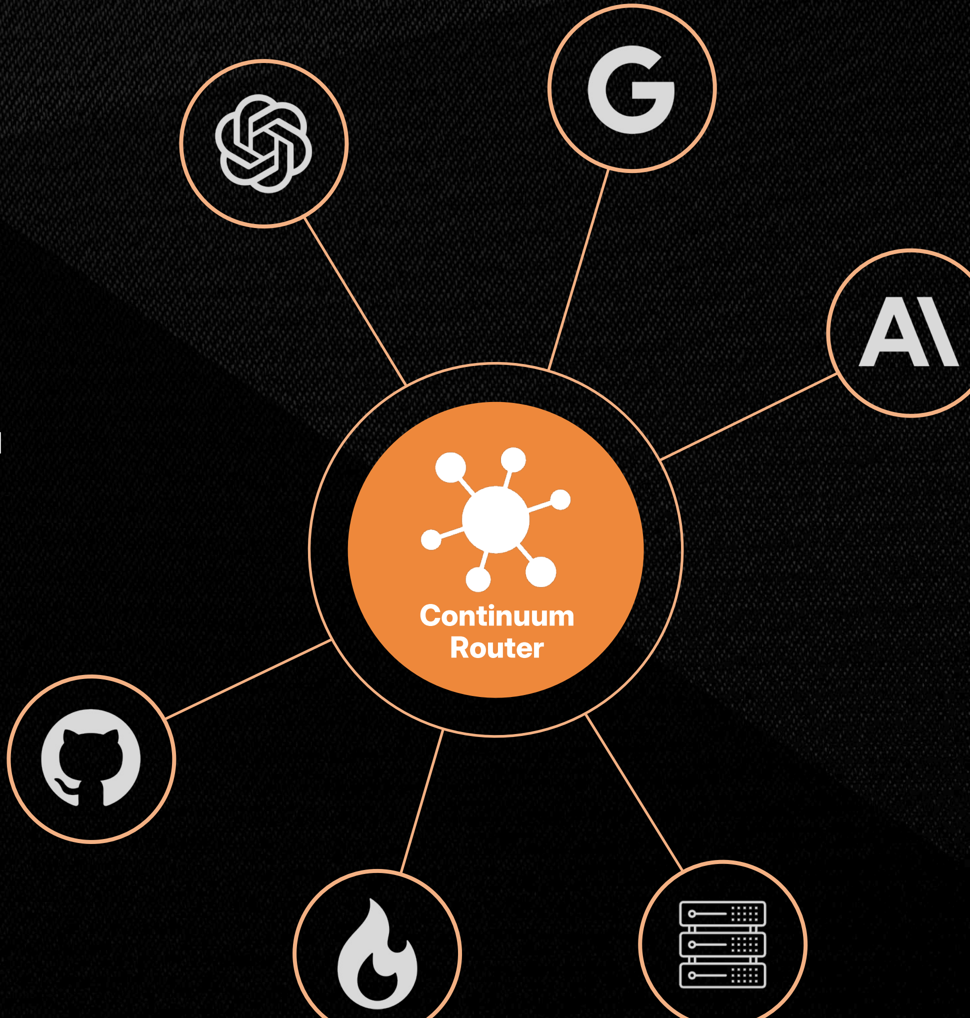
More efficient integration than connecting multiple LLM APIs directly
Seamless integration with existing eco, without code changes through OpenAI-compatible Continuum Router

Enterprise-grade stability for production environments
Automated health checks, circuit breakers, and model fallback capabilities ensure fault isolation and reliability

Unmatched performance
Hot-reload for real-time config changes, advanced routing strategies, and performance with sub-5ms overhead

Complete API ecosystem beyond simple proxying
Integrated essentials for LLM apps including Files API, auto-injection, model discovery, and SSE streaming

Operator-friendly control through Admin REST API
Programmatic interface for managing all aspects of the router through scripts



Continuum suite

Continuum Hub

Enterprise-level LLM monitoring and settlement control plane

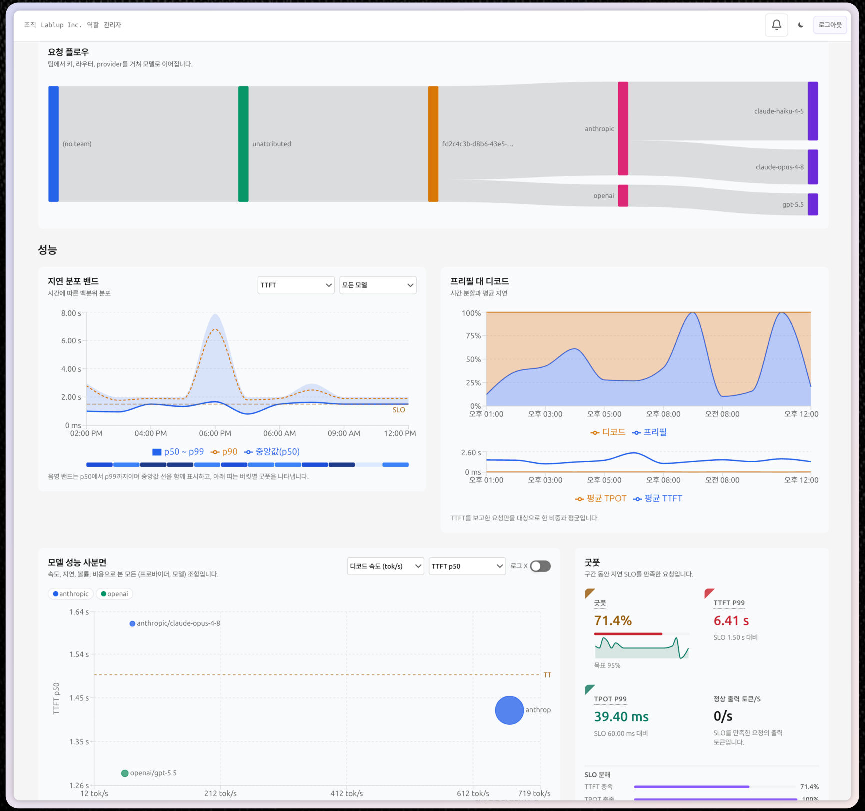
Secure control operating outside the request path
Structure where the router connects only in the outward direction

Usage metering that does not collect body text
Collects only numerical data (metadata), such as token counts and costs, and does not store the body

Organizational control via Keys, Tiers, and Budgets
API keys by team and service, tier limits, automatic blocking upon reaching monthly budget, etc

From weighing to billing, all within a single product
Providing an integrated entire settlement process, including credits, monthly invoice issuance, etc

Manage multiple routers on a single hub
Supports operating multiple sites as a fleet through registration, status reporting, and inspection modes, etc



Product portfolio

AI:GO Generative On-device

Powerful agentic AI platform on your local PCs & devices

Privacy-first design
All data managed within the user's PC without cloud connectivity required for local inference

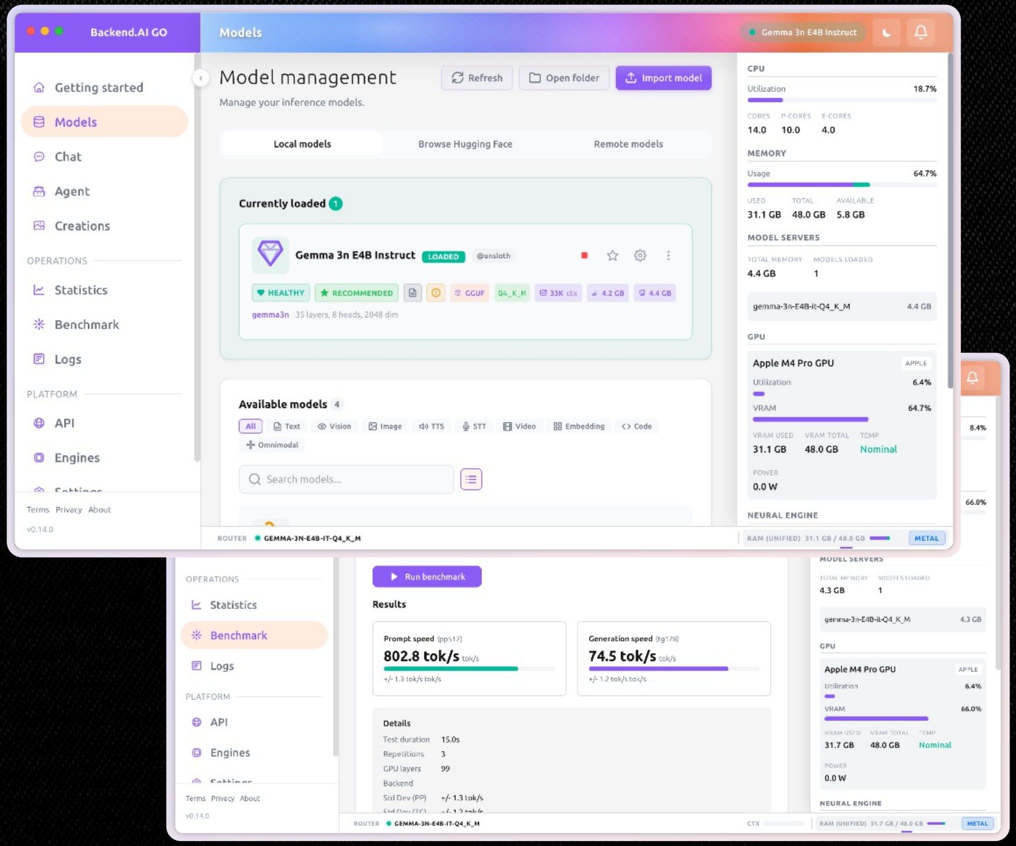
Foundation engine optimized for user hardware
High performance on macOS (Metal/MLX), Windows & Linux (CUDA, ROCm, oneAPI), and CPUs

Open-source first, released as open-source
Search and download Hugging Face models directly within AI:GO, an open-source application

Widely extensible backend through standard APIs
Use AI:GO as the backend for your favorite AI tools via OpenAI-compatible API

Connect and use multiple Backend.AI:GO instances
Run large models on high-spec PCs and connect from edge devices through Mesh networking

Broader reach with Backend.AI and Backend.AI:GO
Utilize existing inference on Backend.AI clusters or launch new inference sessions from Backend.AI



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Track Challenge

Lablup: Making AI Accessible where it wasn't.

JUNCTION Korea 2026

HACK OUR ORIGIN

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Track challenge

Build the Ultimate Agent Squad

Background

The best AI models available today run on massive infrastructure. Not everyone can access the services that frontier model developers offer through APIs built on infrastructure owned by the biggest technology companies. That is why Lablup and FuriosaAI are asking you a question: How far can you go with a model you can actually hold in your hands?

Challenge

For this challenge, you are going to craft two types of output:

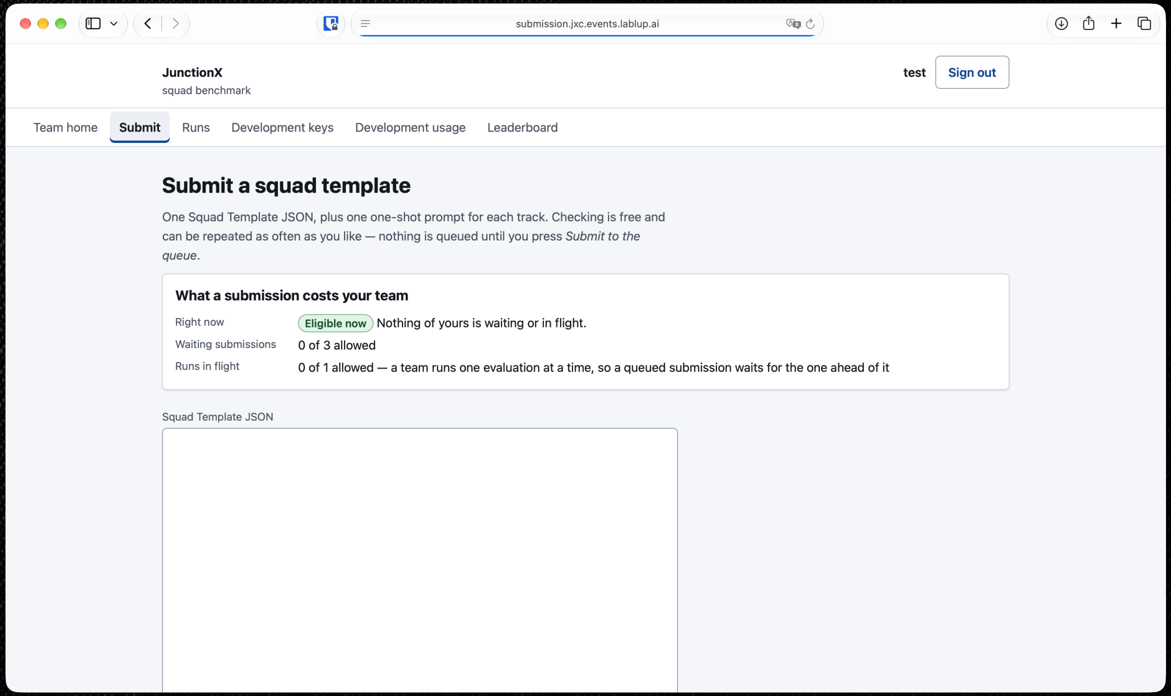
- Problem-solving AI agent squad formation
- Interactive visualization (web, animation generator, etc.) of squad's problem-solving processes.

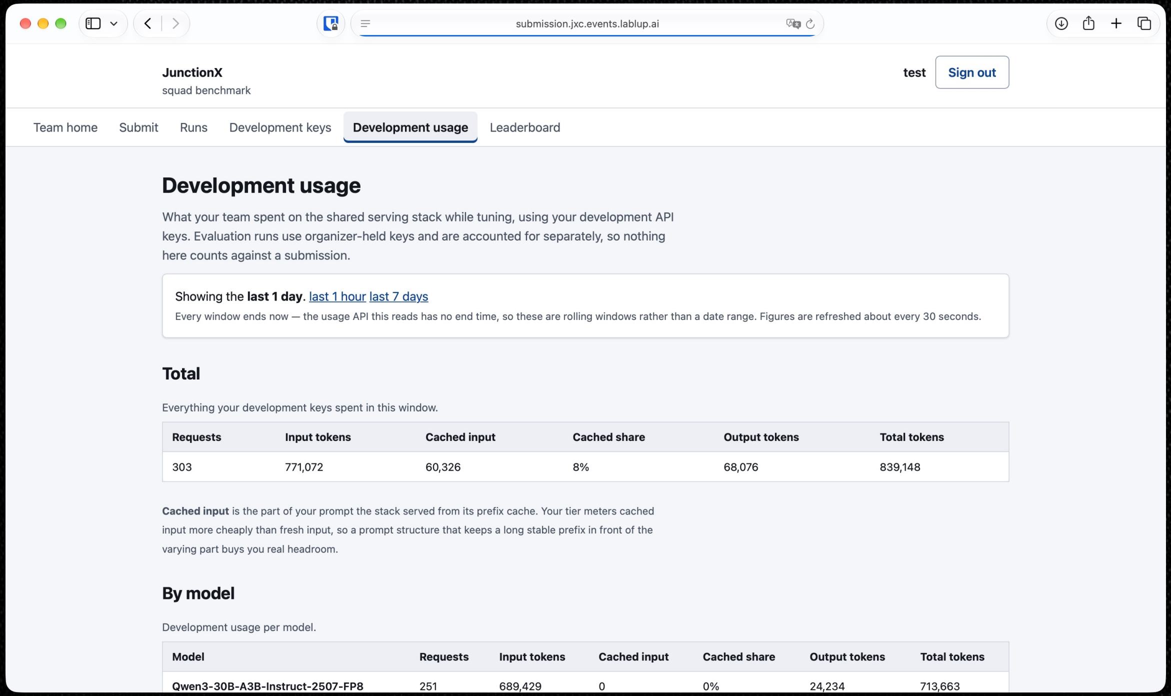
For the former product, your goal is to create and tune an AI agent solving various problems across multiple categories, including but not limited to mathematic, coding and more. You will design the squad entirely on your own, deciding which agents read the code, which ones modify it, which ones verify the results, and which ones determine when it is time to give up. The critical parts of the problem-solving process must be built on AI:GO's agent squad functionality.

Track challenge

Monitor usage, Run test & Submit

https://submission.jxc.events.lablup.ai:8444





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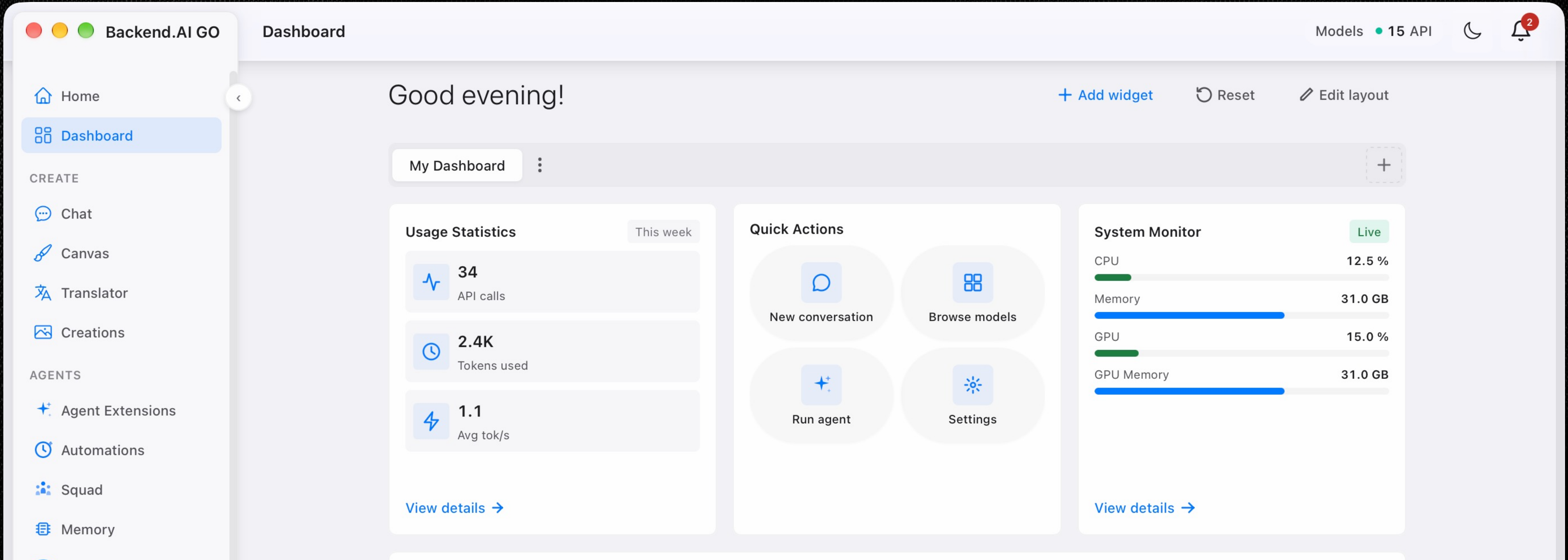
Track challenge

What the heck is the AI:GO?

Your Personal Agentic AI Platform built by Lablup



Track challenge

For the challengers... who are not familiar with AI

We have prepared two short onboarding session for participants.

21:00 – 21:30	lablup	Let them do it: Building your own agent with AI:GO
21:30 – 22:00	FURIOSA	Building the future of sustainable, programmable AI infrastructure

Track challenge

Evaluation & Scoring

Benchmark score (40)	Scores will be awarded in order of the teams' rankings based on their results in the final benchmark test.
Visualization (30)	Scores will be awarded based on how effectively the Trace, represented as log (text) data, is visualized with consideration for observability, interpretability, traceability, explainability, clarity, and insightfulness.
Token efficiency (30)	A reference cost in dollars per million tokens (USD/Mtok) will be set for each model. Teams will be ranked by their total normalized usage cost, with test runs charged at one-fifth of the reference rate and submission runs charged at the full rate. Teams may submit multiple times, and the cost of every submission, including evaluation submissions, will be included. Teams with lower total usage costs will receive higher scores.

Token efficiency

Method and explanation

- **Validation submissions:** Every run is billed at the full rate. If a team runs a submission once, one run is charged; if it runs the submission 100 times, all 100 runs are charged.
- **AI:GO test runs:** Test runs are not fully billed. Only one-fifth of the total cost is counted toward token efficiency.
- **Usage tracking:** Teams can check their total token usage in the Submission Dashboard.
- **Real-time leaderboard:** Benchmark results submitted by all participants are publicly displayed.
- **Run-level details:** Participants can view the final score for each benchmark run, along with the number of tokens used by each model during that run.

Teams that achieve comparable benchmark results with lower token usage will receive a higher token-efficiency score.

Prize and benefits

- **#1 Winning team** KRW 2,000,000 + Lablup Merchandise (Worth KRW 50,000, per member)
- **#2 Winning team** KRW 1,000,000 + Lablup Merchandise (Worth KRW 50,000, per member)

- **Winners (All)** Lablup company tour & Coffee/career-chat opportunities (for those who are interested)

“Make AI Accessible”

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"Make AI Scalable"

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Democratizing intelligence for humanity.

MAIL	contact@lablup.com
Lablup Inc.	https://www.lablup.com
Backend.AI	https://www.backend.ai
GitHub	https://github.com/lablup/backend.ai
Cloud	https://cloud.backend.ai